

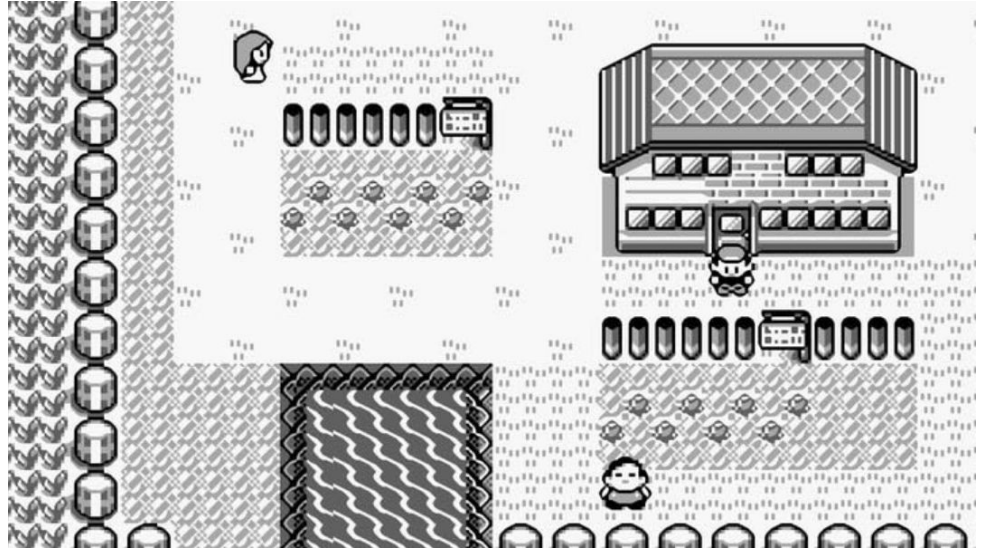
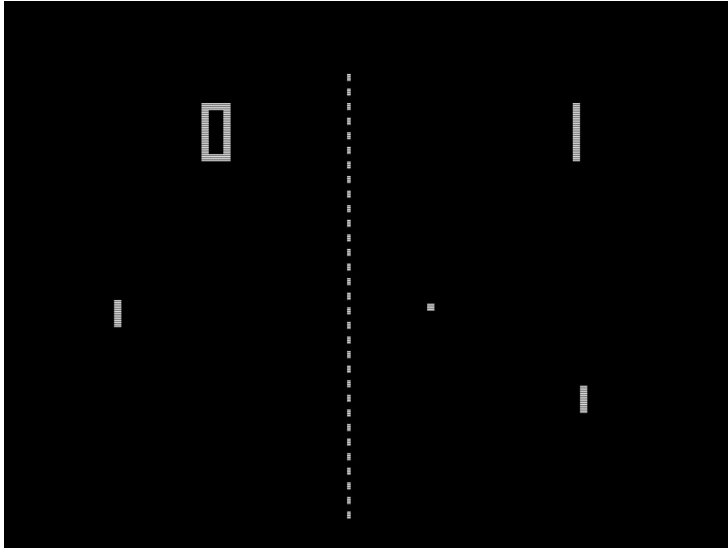
Colour & Constraint

Designing retro game palettes

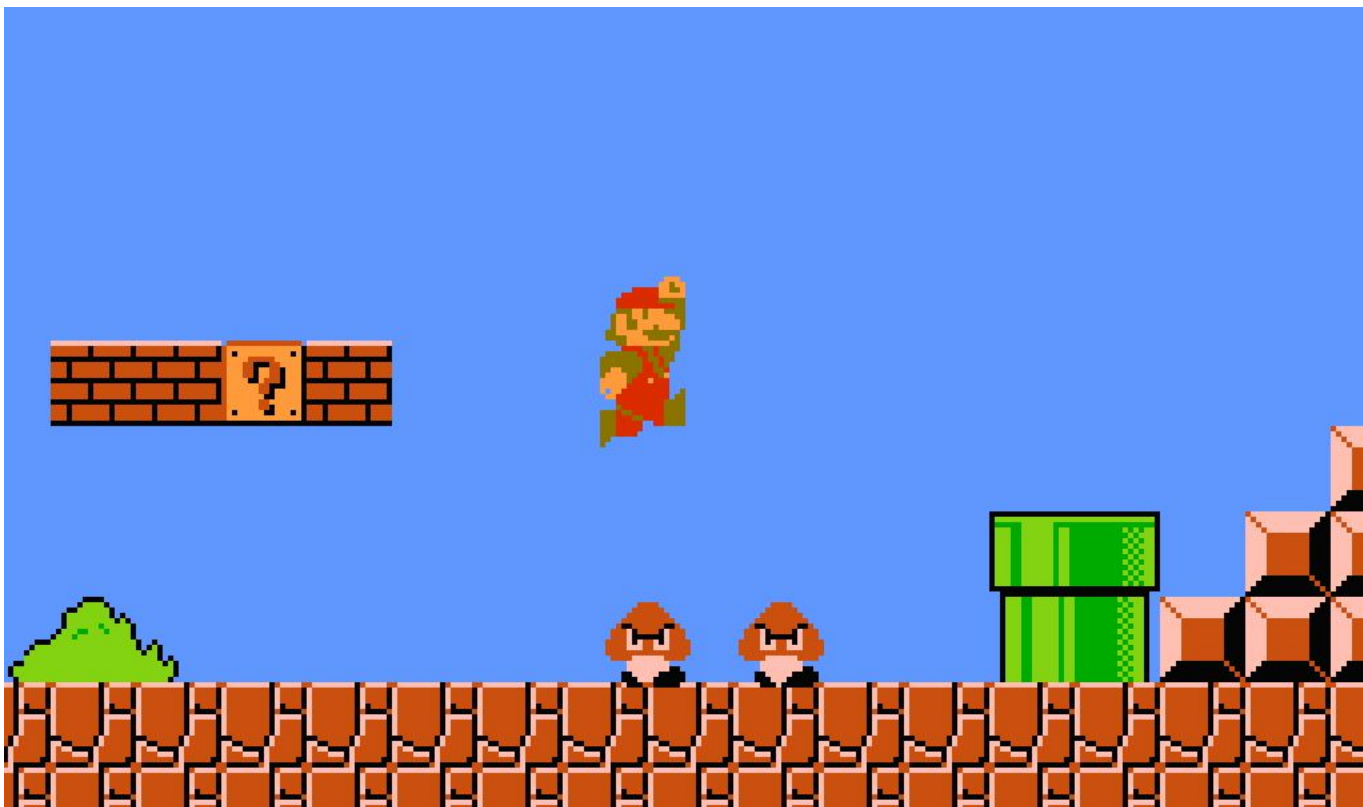
Part 1: Colour in retro games

What are Retro games again?

Early games - Monochrome



Retro games - 8-bit



Retro games - 8-bit



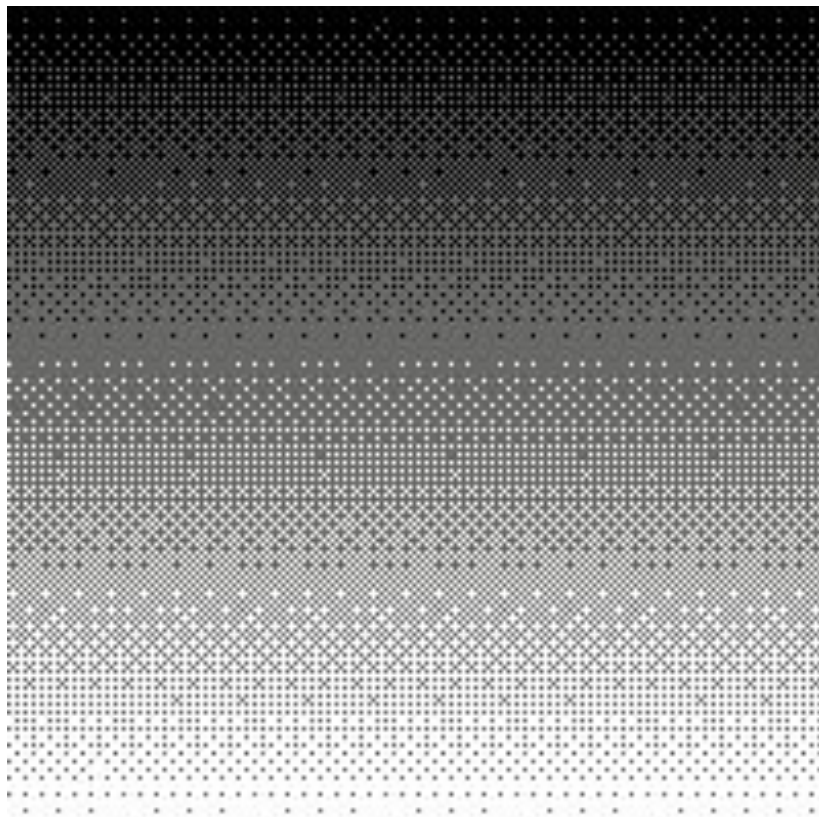
16-bit



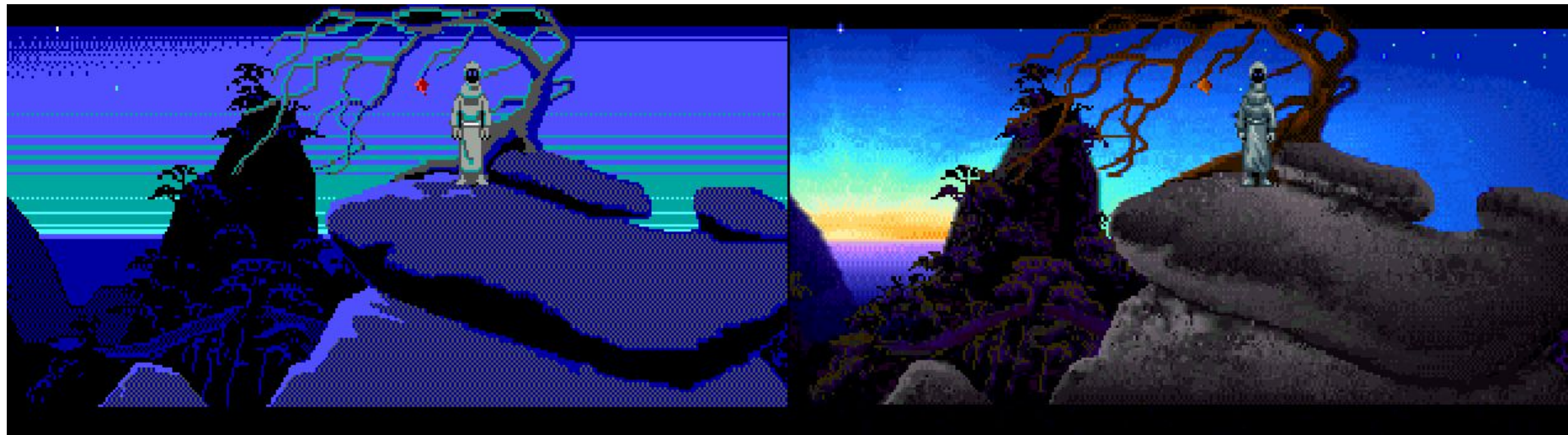
Dithering



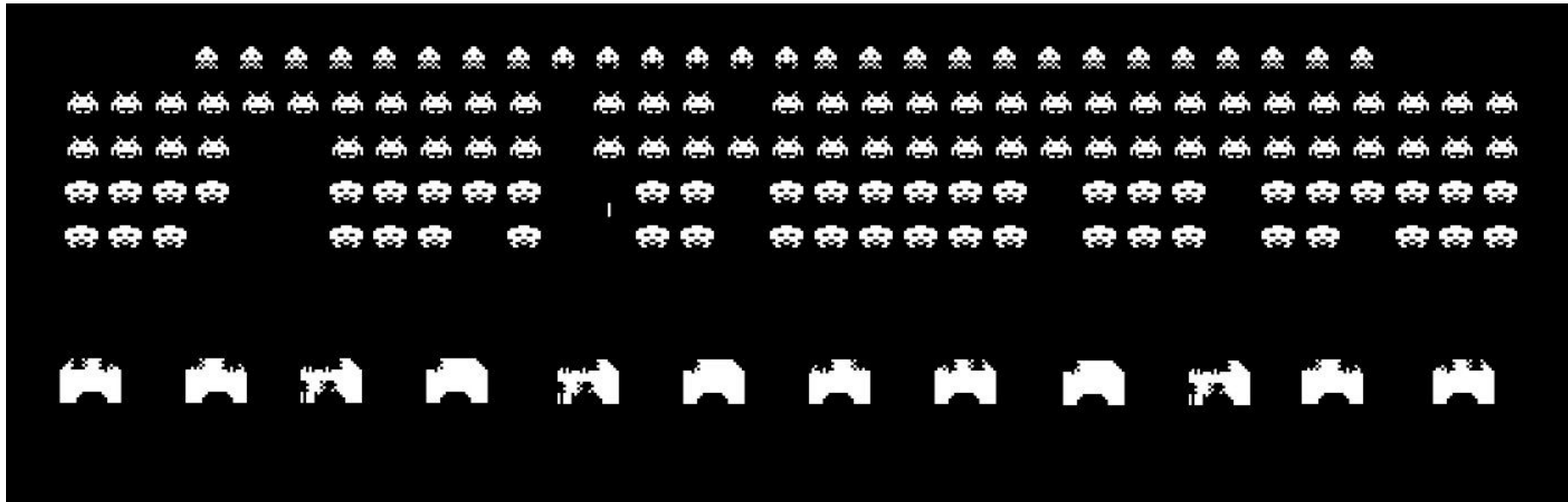
Dithering



Dithering



Why even develop colour?



Why even develop colour?



Retro revival and displaying colour



Part 2: Colour theory

Basis of colour theory - definitions

hue/saturation/value/palette/scheme

Hue



Saturation



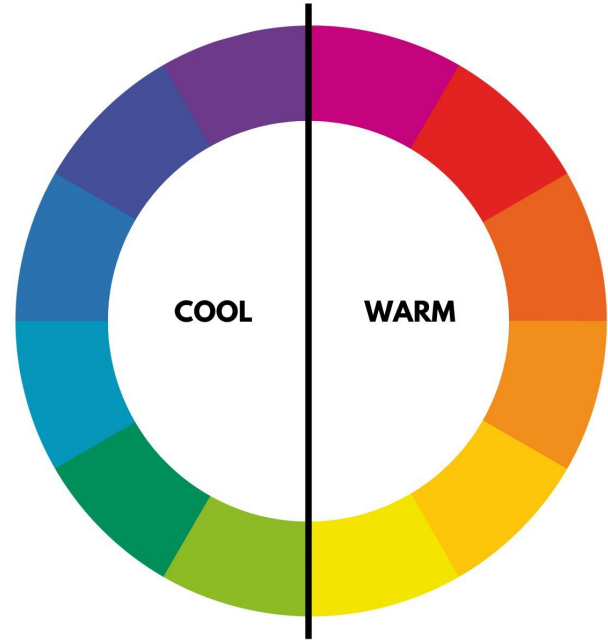
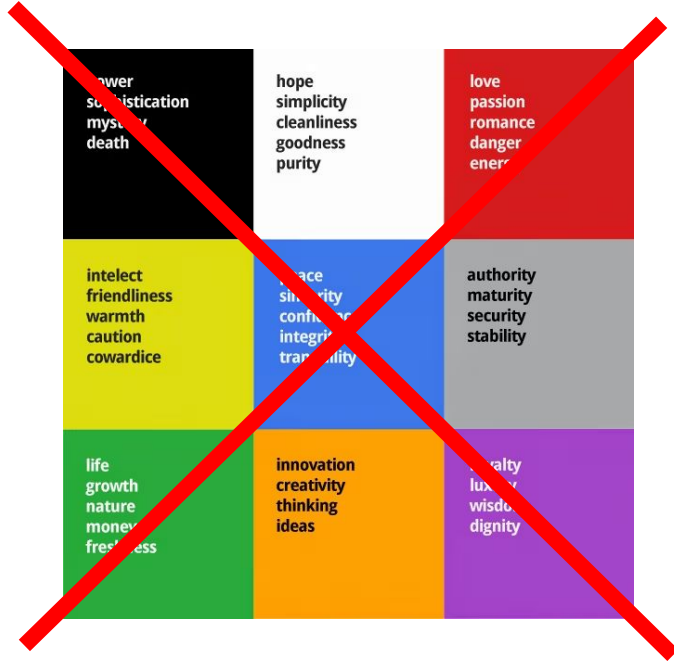
Brightness



Basis of colour theory - colourwheel



Basis of colour theory - Important cultural note (warm/cool)



Basis of colour theory - Important cultural note example



Orange shirt day, Canada



Kings Day, the Netherlands

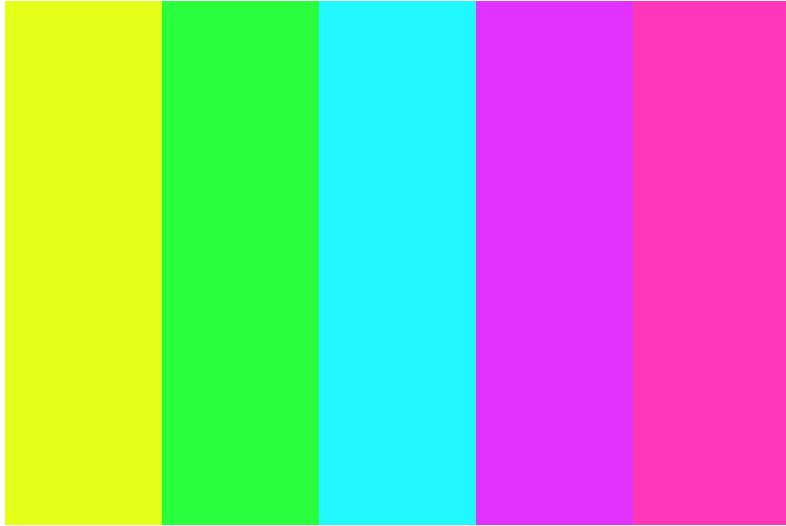


Buddhist monks, India

All wear orange, but with completely different meanings and associations

Basis of colour theory - some universal anchors

Visual hierarchy, harmony & strain

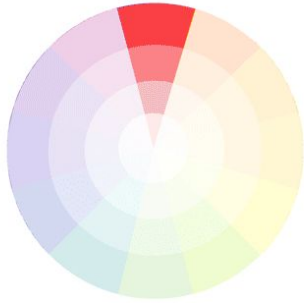


Large fields of saturated colours
are very straining



Keep in mind the visual hierarchy,
what do you need to show first

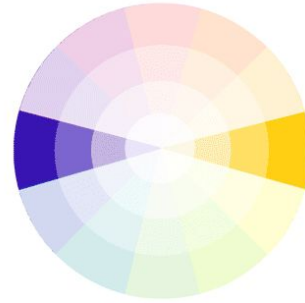
Colour theory schemes - overview



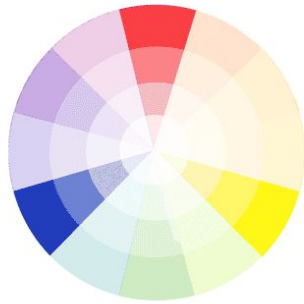
Monochromatic



Analogous



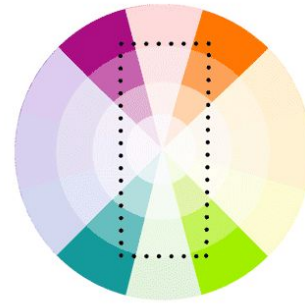
Complementary



Triad

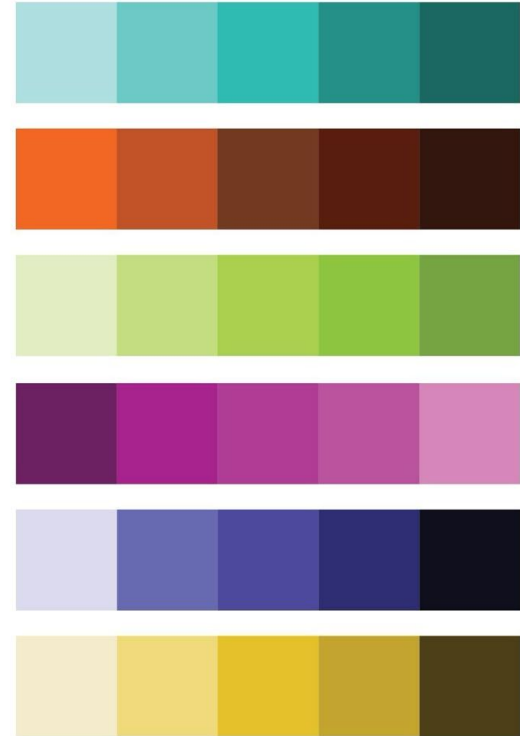
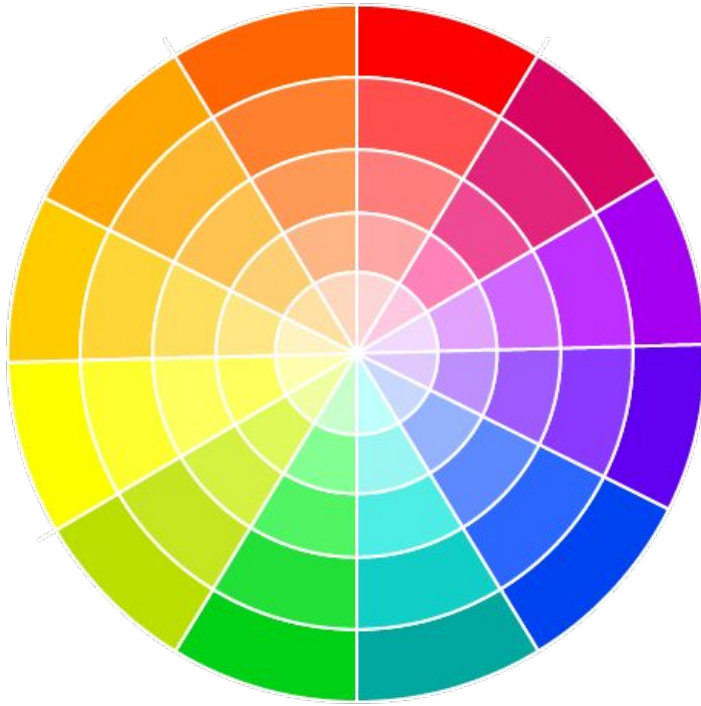


Split-Complementary

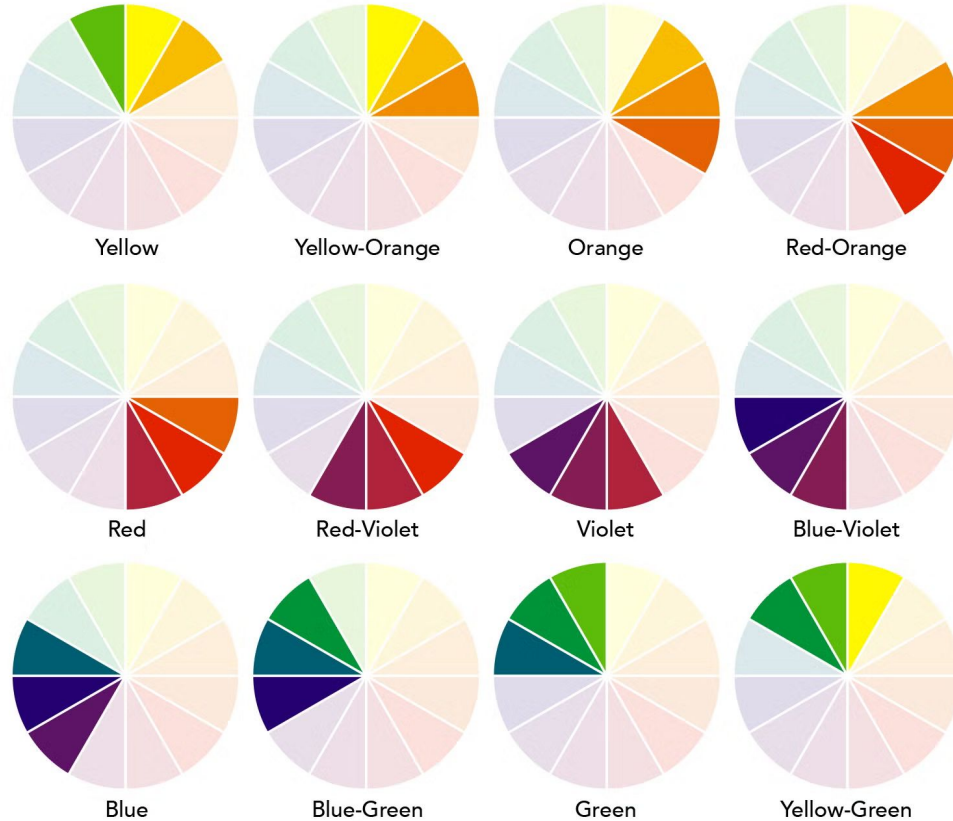


Tetradic

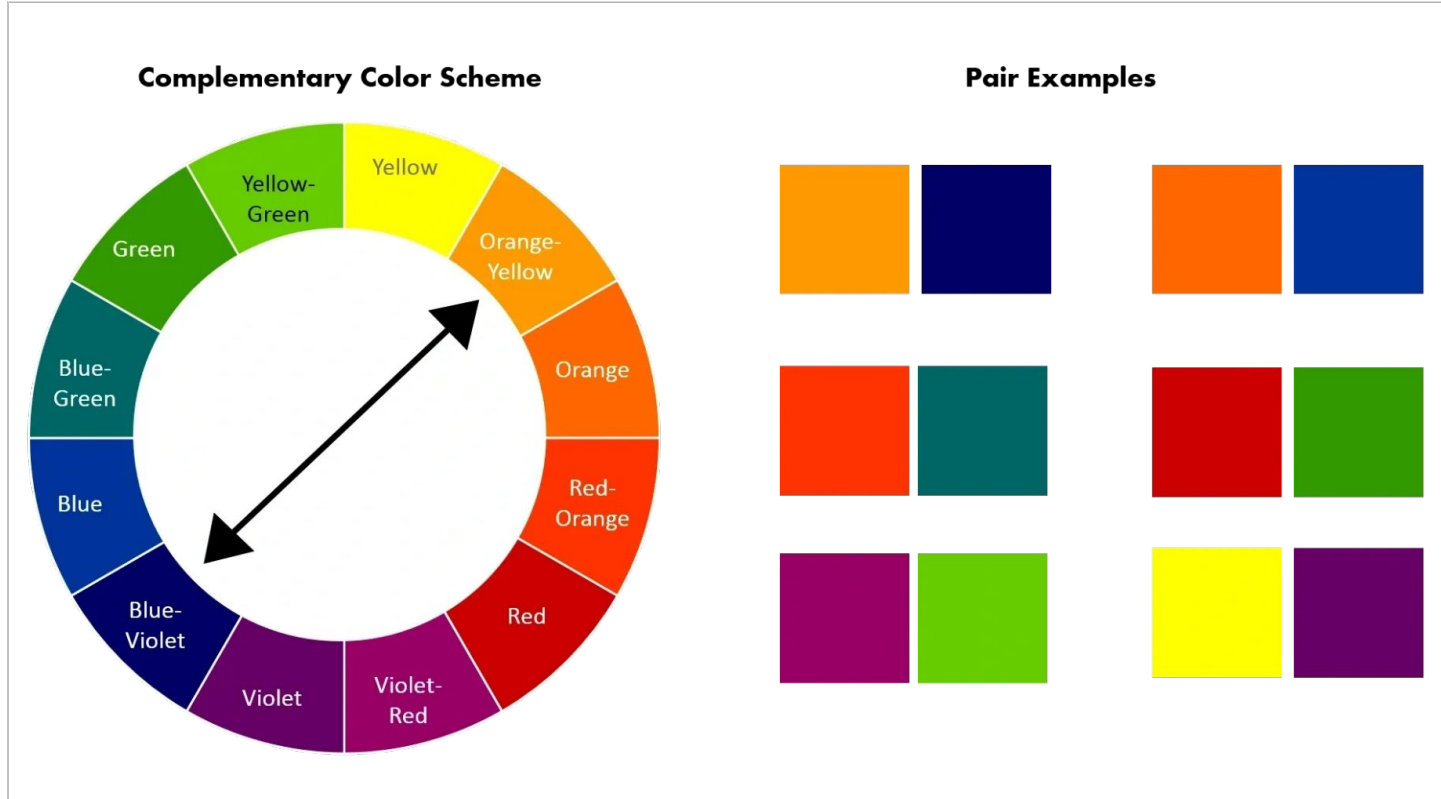
Colour theory schemes - monochrome



Colour theory schemes - analogous



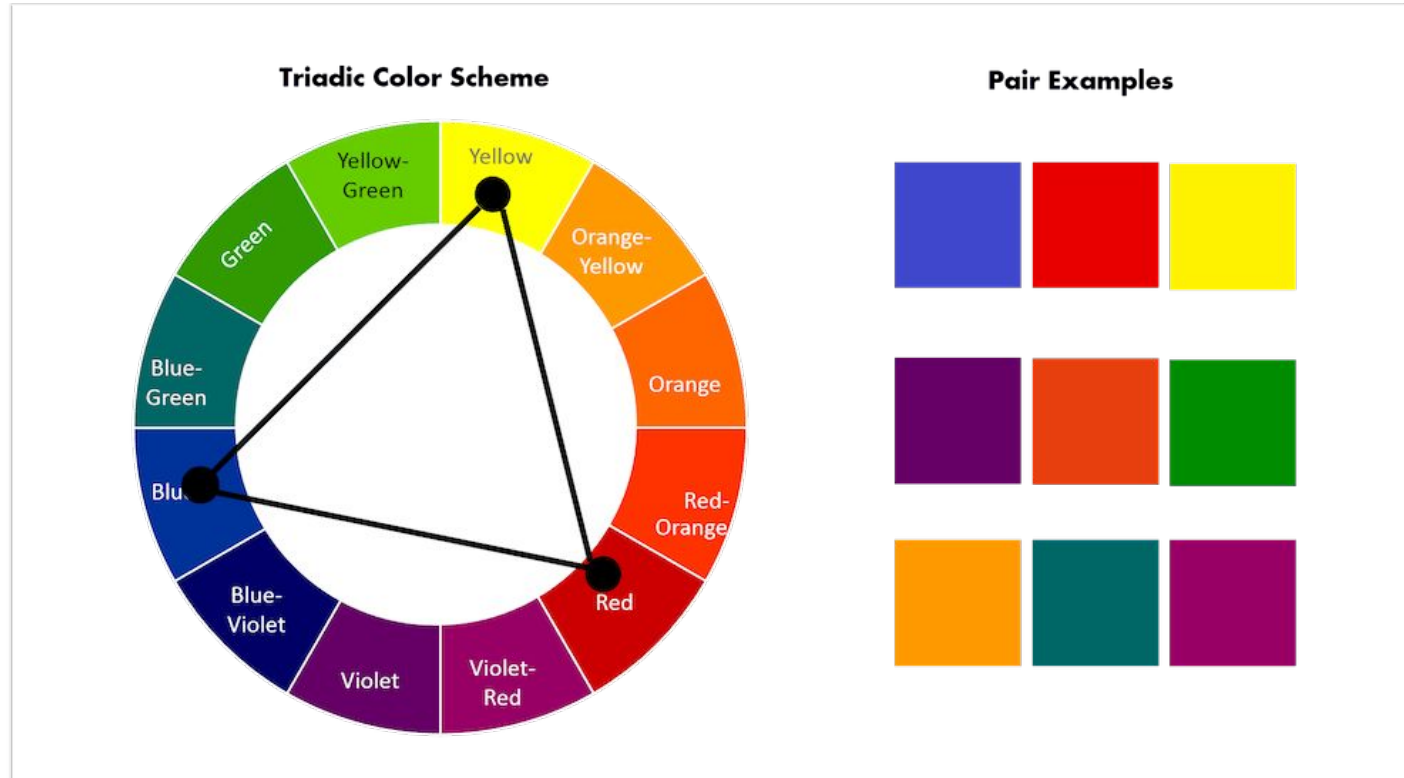
Colour theory schemes - complementary



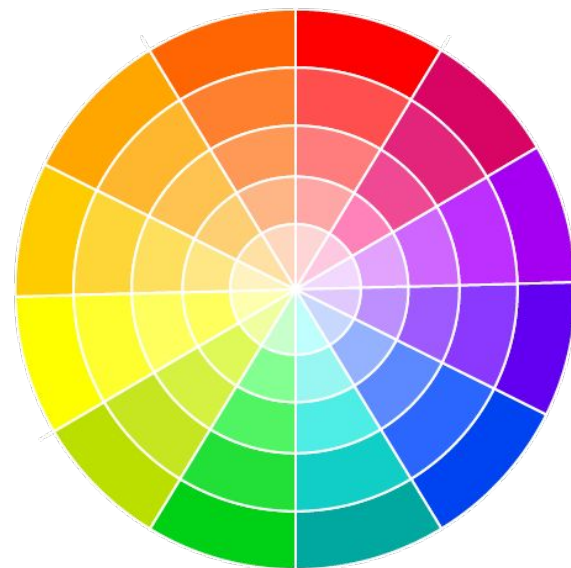
Colour theory schemes - split complementary



Colour wheel schemes - triadic



Colour wheel schemes - Square



Part 3: How to colour things?

Practical guide to picking a palette - overview

Picking a palette - step 1: pick a mood

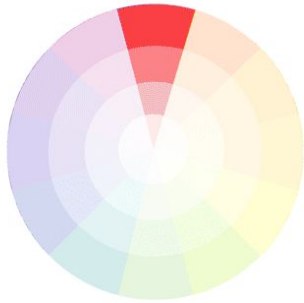
Low energy / High energy => low saturation / high saturation

Subtle / Loud => low contrasts / high contrasts

Harmonious / Conflicting => analogous / complementary

Etc. etc. etc.

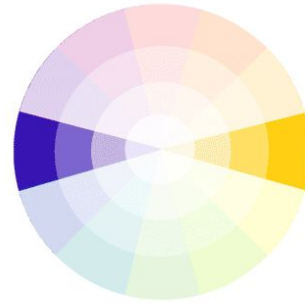
Picking a palette - Step 2: Take stock & pick a scheme



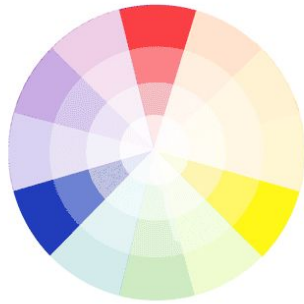
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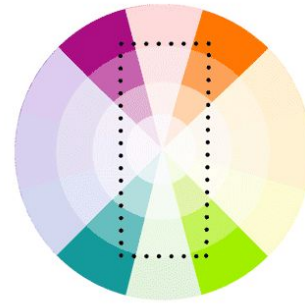
Complementary



Triad

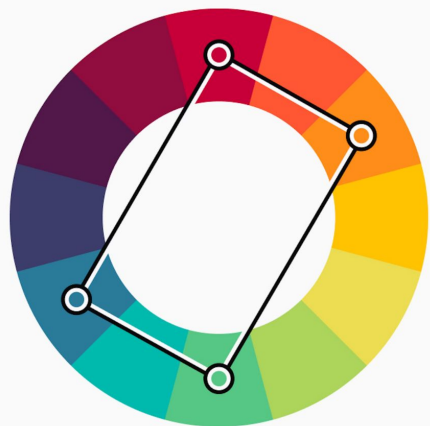


Split-Complementary



Tetradic

Picking a palette - step 3: pick a palette



Red / Orange / Green / Blue



Orange / Yellow / Blue / Purple



Yellow / Green / Purple / Red



Vermilion / Amber / Teal / Violet



Amber / Chartreuse / Violet / Magenta



Chartreuse / Teal / Magenta / Vermilion



TRIADIC COLOR COMBINATIONS



But wait, wasn't the whole point of the triadic scheme to pick **THREE** colours?

YES, you still see three colours; red, yellow and blue. You are allowed to add brighter/darker shades of the same colour.

Picking a palette - step 4: check the contrast

Good		Good	
Bad		Bad	
Good		Good	
Bad		Bad	
Good		Good	
Bad		Bad	

Picking a palette - Tip 1: reuse aggressively

Keep using your colour palette to create a sense of consistency and continuity.

Picking a palette - Tip 2: use accents carefully

